

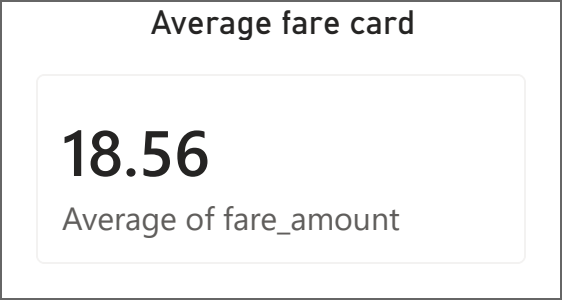
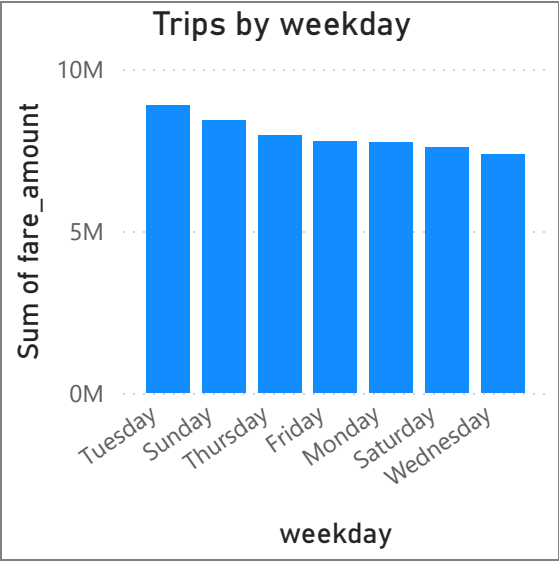
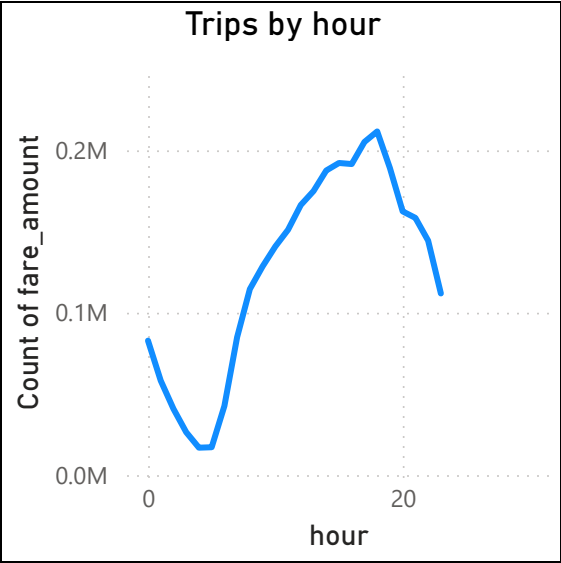
Data Trust & Cleaning



Invalid trips (zero distance, negative fares, incorrect timestamps) were removed to ensure reliable analysis.

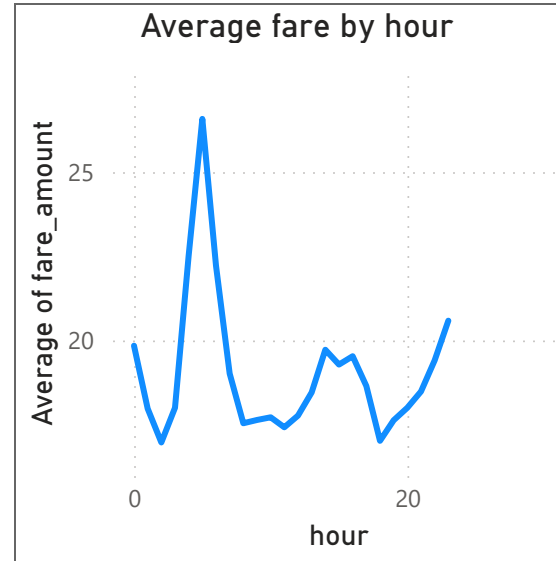
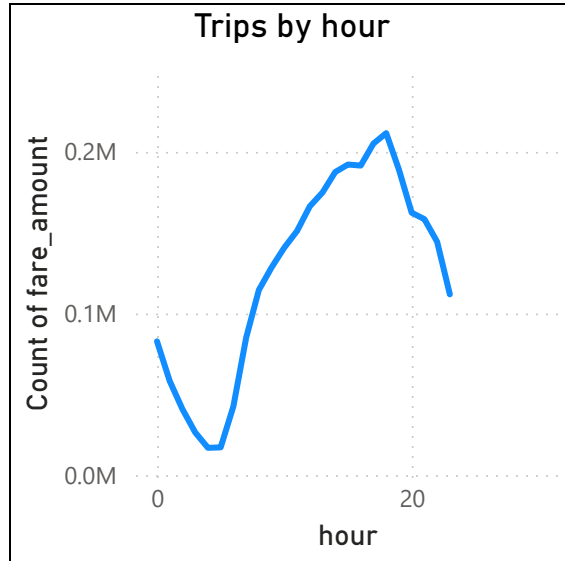
Sample Trip Records							
Year	Quarter	Month	Day	Count of trip_distance	Count of fare_amount	Count of tip_amount	Count of payment_type
2008	Qtr 4	December	31	1	1	1	1
2022	Qtr 4	October	25	6	6	6	6
2022	Qtr 4	December	31	25	25	25	25
2023	Qtr 1	January	1	74474	74474	74474	74474
2023	Qtr 1	January	2	63928	63928	63928	63928
2023	Qtr 1	January	3	83676	83676	83676	83676
2023	Qtr 1	January	4	92959	92959	92959	92959
2023	Qtr 1	January	5	98816	98816	98816	98816
2023	Qtr 1	January	6	100331	100331	100331	100331
Total				2998379	2998379	2998379	2998379

Observed System Behavior



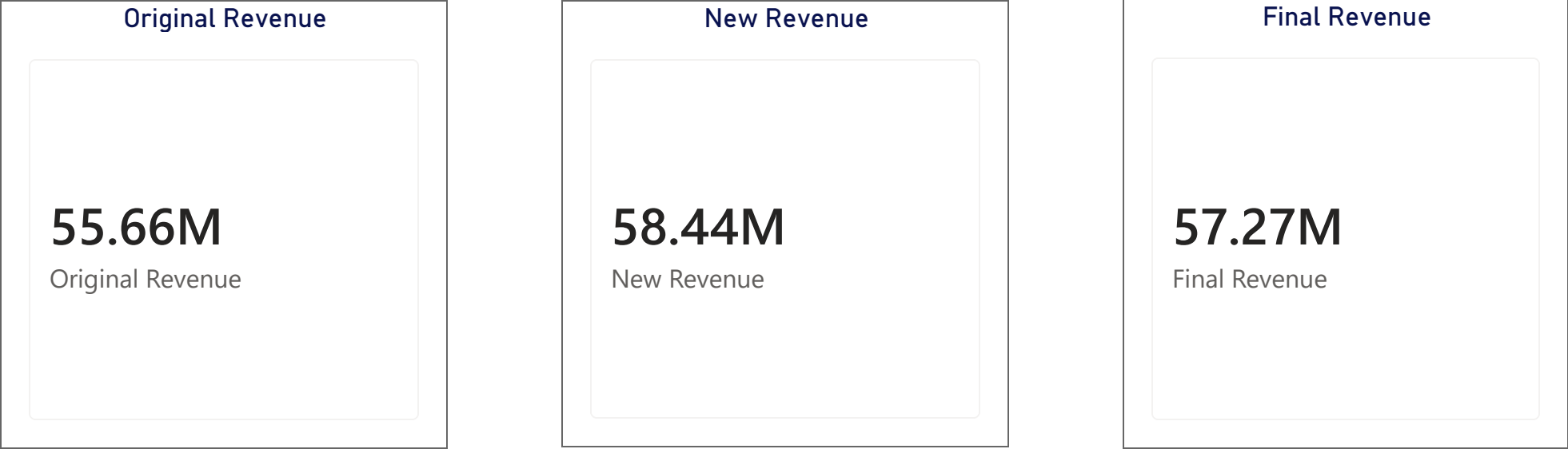
Trips peak in evening hours and drop late night, reflecting real commuter behavior.

System Stability & Strain



Peak hours indicate system pressure (high demand).
Low activity periods show stability.
The system follows a predictable daily pattern.

Policy Impact Simulation



A 5% fare increase raises revenue.
However, a 2% drop in demand reduces the gain.
The system absorbs the change rather than amplifying it.

If demand drops further, revenue could decrease.
System behavior depends on passenger response.

Wrong: Higher fares always increase revenue.

Correct: Higher fares can reduce demand, limiting revenue gains.

Wrong: More trips = system efficiency.

Correct: Peak demand can indicate system strain.